PROJECT NAME :

A MINI COMPUTER DESIGN

SUBMITTED TO:

M.M.A. Hashem
Professor
Department of Computer Science & Engineering
Khulna University of Engineering & Technology (KUET)
Khulna -9203, Bangladesh.

Md. Badiuzzaman Shuvo
Lecturer
Department of Computer Science and Engineering
Khulna University of Engineering & Technology (KUET)
Khulna -9203, Bangladesh.

SUBMITTED BY:

Mehedi Hasan Rabby
Roll:2107086
Department of Computer Science and Engineering
Year: 2nd   Term: 1st
Course Title: Computer Architecture
Course No: CSE-2114
Bit: 18

Objectives:

1. To learn about how to implement an Arithmetic and Logic Unit.
2. To learn about how to implement a Control Unit of a computer.
3. To learn about how to implement a Central Processing Unit .
4. To learn about how to add different kinds of instruction on CPU.

Introduction:

An 18-bit ALU is a combinational digital circuit that performs various arithmetic and logical operations on binary data inputs. It typically includes operations such as addition, subtraction, multiplication, division, logical AND, logical OR, logical XOR, and bit shifts. The 18-bit ALU I built can perform these operations on 18-bit binary numbers.

The CPU, on the other hand, is a more complex digital circuit that serves as the central processing unit of a computer system. It is responsible for fetching instructions from memory, decoding them, executing them, and managing data flow between various components of the computer system.

In Logisim, I have designed the CPU using various digital logic components such as registers, counters, multiplexers, decoders, and control units. The CPU have the following main components:

Instruction Register (IR): This register holds the current instruction fetched from memory. It stores the opcode (operation code) that tells the CPU what operation to perform and any operands or addressing modes required by the instruction.

Program Counter (PC): The program counter keeps track of the memory address of the next instruction to be fetched. After fetching an instruction, the PC is typically incremented to point to the next instruction in memory, allowing the CPU to proceed with program execution sequentially.

Arithmetic Logic Unit (ALU): The ALU is responsible for performing arithmetic and logical operations on binary data. It can perform operations such as addition,

subtraction, multiplication, division, logical AND, logical OR, logical XOR, and bit shifts. In your case, it's designed to handle 18-bit binary numbers.
Accumulator (AC): The accumulator is a special register used to store the result of ALU operations. After the ALU performs a computation, the result is often stored in the accumulator for further processing or to be moved to another location in memory.
Control Unit: The control unit generates control signals that coordinate the operation of other CPU components based on the instruction being executed. It interprets the opcode from the instruction register and generates signals to control the ALU, memory access, and other operations required to execute the instruction.
Memory (RAM and ROM): Memory stores both instructions and data. Random Access Memory (RAM) is used for storing data and variables that may change during program execution, while Read-Only Memory (ROM) typically stores the program instructions that do not change during normal operation, such as the program's initial boot sequence.

The CPU fetches instructions from memory by using the program counter to access memory addresses sequentially. Once fetched, the instruction is stored in the instruction register, and the control unit decodes the instruction to determine the operation to be performed. Based on this decoding, the control unit generates signals to control the ALU, memory access, and other components as needed to execute the instruction.

Sample Program:

Instruction: Hex code

1.Load R1,[0Ch]----------------------0060c
2.Add [0dh]-----------------------------0010d
3.Ex-Or[0eh]------------------------------0080e
4.Not[00]-----------------------------------00900
5.Halt[00]-----------------------------------00700

Discussion and Conclusion:

The design and implementation of a minimal computer system in Logisim provided valuable insights into the inner workings of computer architecture. Through this project, I gained practical experience in designing and integrating key components such as ALU, registers, control unit, and CPU, as well as understanding their interconnections and interactions.

The project highlighted the importance of a systematic design approach, starting from basic building blocks and gradually expanding to more complex components. By simulating the operation of a computer system at the digital logic level, this project helped understanding concepts such as instruction execution, data manipulation, and control flow.

Overall, this project served as an educational exercise in computer architecture equipped with foundational knowledge and practical skills essential for further exploration and study in the field of computer science and engineering.